Salmonella Enteritidis infection, corticosterone levels, performance and egg quality in laying hens submitted to different methods of molting

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ABSTRACT In commercial layer poultry farming, molt induction is an important tool used by egg producers to prolong the production cycle of laying hens. Conventional molt induction programs involve total feed withdrawal, which raises questions about animal welfare and increased infection susceptibility. The high incidence of paratyphoid salmonellosis infections in commercial poultry farming is still an important health challenge because in addition to affecting the birds, such infections also cause public health problems. In this context, experiments were performed with laying hens at 79 wk of age to compare the conventional forced molting method (fasting) with an alternative method (free wheat bran supply) and determine their effect on the persistence of vaccine antibodies against Newcastle disease, the control and reduction of experimentally inoculated *Salmonella* Enteritidis, and the performance and egg quality of hens. A reduction ($P < 0.05$) of *Salmonella* Enteritidis in the crop and lower production of corticosterone were observed in the birds that received wheat bran compared with those subjected to total fasting. Moreover, a better performance ($P < 0.05$) with regard to egg production, egg mass, and feed conversion/kg and dozen eggs was observed in the hens that received the alternative treatment compared to the conventional forced molting method. Thus, the use of wheat bran for forced molting was found to be feasible and met the welfare needs of the hens.

Key words: molt induction, *Salmonella* Enteritidis, corticosterone, egg quality, laying hens

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INTRODUCTION

Under natural conditions, feather molting is a natural physiological process in birds. In commercial layer poultry farming, molt induction is an important tool commonly used by egg producers to prolong the production cycle of laying hens, which results in significant weight loss, dynamic changes in the morphology and physiology of the reproductive system, interrupted egg production for a temporary period, satisfactory improvements in egg quality, and maximized post-molt egg production (Cunningham and Mauldin, 1996; Berry, 2003; Webster, 2003; Butcher and Miles, 2011; Jeong et al., 2013; Flock and Anderson, 2016).

Conventional molt induction programs involve photoperiod reductions and total feed withdrawal (Hembree et al., 1980; Davis et al., 2002); however, such methods have been criticized because of the effects on animal welfare (Keshavarz and Quimby, 2002; Bell, 2003). Moreover, some researchers also raise concerns about food safety and increased infection susceptibility due to stress, which raises the plasma concentrations of corticosterone and other hormones and reduces the immune response to certain diseases, such as salmonellosis and Newcastle disease (Davis et al., 2000; Holt, 2003; Ricke, 2003; Sandhu et al., 2007). A well-known force molting program, called California program, incorporates a period of complete feed withdrawal followed by nothing but whole milo, wheat or cracked corn for a period (North, 1984).

Bacteria of the genus *Salmonella* are often reported in layer poultry farming and in human food outbreaks involving the consumption of eggs and their contaminated by-products worldwide (Foley et al., 2011). The implementation of management practices resulting in reduced stress in laying hen flocks is an accepted practice to minimize outbreaks of salmonellosis because stress is often the catalyst for clinical outbreaks in laying hens and broiler chickens (Butcher and Miles, 2011). In particular, infection by serovar Enteritidis has been extensively studied during the molting period (Holt, 2003; Ricke, 2003; McReynolds et al., 2005, 2006), and it has been proposed that alternative feed supply during molting periods can promote changes in the gastrointestinal system, thereby preventing the invasion and establishment of *Salmonella* Enteritidis (Seo et al., 2002).

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Several studies have investigated and improved upon alternative molting methods to provide birds with better conditions. Diet manipulation with high wheat bran contents is less aggressive and has been mentioned as a potential alternative, and its results are comparable to that of conventional molt induction methods (Biggs et al., 2003, 2004; Khoshoei and Khajali, 2006). Therefore, the objective of the present study was to compare the conventional forced molting method (fasting) with an alternative method (free wheat bran supply) to determine the effects of these methods on the persistence of vaccine antibodies against Newcastle disease, production of corticosterone, effectiveness in controlling and reducing infection by *Salmonella* Enteritidis, and performance and egg quality of hens.

MATERIALS AND METHODS

Experiment I

The experiments were conducted at the Experimental Laboratory of Avian Pathology, Department of Clinical Veterinary Science, School of Veterinary Medicine and Animal Science—FMVZ, Sao Paulo State University—UNESP, Botucatu Campus, Sao Paulo, Brazil. The experimental protocols were submitted and approved by the Committee on the Ethical Use of Animals—CEUA—UNESP, and they complied with the ethical principles in animal experimentation adopted by the Brazilian College of Animal Experimentation—COBEA (CEUA n.74/2014).

Birds were euthanized via the administration of 2 mg/kg xylazine intramuscularly followed by administration of 15 mg/kg thiopental sodium intravenously. After bird unconsciousness was detected, 3 mL of potassium chloride (KCl) was administered to induce death (American Veterinary Medical Association, 2007). All necessary precautions were taken to reduce animal suffering during the study.

Salmonella Enteritidis Strain and Experimental Challenge

As an experimental challenge, the *Salmonella enterica* subspecies *enterica* serovar Enteritidis strain, which is resistant to nalidixic acid (Nal–100 μg/mL of medium) and rifampicin (Rif–100 μg/mL of medium) and belongs to the bacterial collection of the Laboratory of Avian Pathology, FMVZ—UNESP, was used. The challenge inoculum consisted of culturing the *Salmonella* Enteritidis strain in brain heart infusion (**BHI**) and incubating at 40°C for 18 h. The number of colony forming units (CFUs) was determined by decimal dilutions in phosphate-buffered saline (PBS) at pH 7.2. All bacterial determinations were performed by plating 0.1 mL of the suspensions (BHI) and respective decimal dilutions (PBS) in duplicate in brilliant green agar (BGA) (Nal/Rif). The plates were read after incubation at 40°C for 24 h.

On the fourth day after the start of forced molting, all birds were administered orally via gavage a 1 mL dose of the *Salmonella* Enteritidis inoculum at a concentration of 10^8 CFU/mL.

Chickens and Facilities

Lohmann LSL-Lite laying hens at 79 wk of age were used. The hens were housed in galvanized wire cages measuring 0.69 m in height, 0.99 m in width, and 0.96 m in length. The birds were randomly distributed into 3 experimental groups, and 4 replicates with 25 birds each were implemented, thus totaling 100 birds per treatment: group 1—wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum; group 2—total feed restriction (California method); and group 3—production feed ad libitum without any type of restriction (control).

Sanitary and Dietary Management

The housing, feeding management, and daily procedures were similar for all hens and based on the guidelines of the Lohmann LSL-Lite line laying hen management guide (Lohmann do Brasil, 2017). The bird vaccination program is shown in Table 1.

The experimental treatment diets were formulated with corn and soybean meal (Table 2), and the nutritional levels followed those recommended by the Lohmann LSL-Lite line laying hen management guide (Lohmann do Brasil, 2017).

To ensure that the birds were free of *Salmonella* spp. at the time of housing, cloacal swabs were collected from each bird and 5 hens were euthanized for bacterial investigation using the method recommended by Mallinson and Snoeyenbos (1989). The feed was also analyzed following the method of the same authors to verify the absence of *Salmonella* spp. before supplying it to the birds.

The birds were subjected to a natural photoperiod throughout the experiment except for the control group, which was exposed to light for 17 h daily. The experiments were conducted during the spring period. Feed and water were supplied ad libitum until the beginning of the experiment.

Experimental Design

The birds were distributed following a completely randomized design into 3 treatments and 4 replicates with 25 birds each. The experiment lasted 10 D, and hens that lost 25% of their weight were fed production feed ad libitum.

Table 1. Vaccination program of commercial laying hens submitted to different types of molt induction programs.

Age (days)	Disease	Via
18 (incubation)	Fowl pox, Marek´s disease and Gumboro disease	*In ovo*
2	Avian pneumovirus	*Spray*
6	Newcastle disease (HB1) and infectious bronchitis (Massachusetts)	Eye-drop
21	*Escherichia coli*	*Spray*
27	Newcastle disease (La Sota), infectious bronchitis (Massachusetts), and Gumboro disease (strong strain Australia V-877)	Drinking water
42	*Mycoplasma Gallisepticum* and infectious coryza (bacterin of *Haemophilus paragallinarum*)	Intramuscular
50	Avian pneumovirus	*Spray*
65	*Mycoplasma gallisepticum* and avian encephalomyelitis	Wing web stab
65	Fowl pox and avian encephalomyelitis	Wing web stab
65	Newcastle disease (La Sota) and infectious bronchitis (Massachusetts)	*Spray*
84	*Escherichia coli*	*Spray*
105	Newcastle disease, infectious bronchitis, egg drop syndrome, and infectious coryza	Intramuscular
180	Newcastle disease (La Sota) and infectious bronchitis (Massachusetts)	*Spray*
270	Newcastle disease (La Sota) and infectious bronchitis (Massachusetts)	*Spray*
360	Newcastle disease (La Sota) and infectious bronchitis (Massachusetts)	*Spray*

Table 2. Composition of basal diets of commercial laying hens submitted to different types of molt induction programs.

	Forced molting (0 to 14 D)		
Ingredients (%)	Wheat bran	California method	Ad libitum feed
Corn	–	–	64.5
Wheat barn	70	–	–
Soybean meal (45%)	–	–	21.8
Meat meal	–	–	3.9
Limestone coarse	30	–	8.9
Sodium bicarbonate	–	–	3.5
DL-Methionine	–	–	0.05
Vitamin[1]/mineral[2] premix	–	–	0.3
Mycotoxin adsorbent			0.1
TOTAL	100	000	100
Calculated nutrient composition			
ME (kcal/kg)	1824	–	2760
CP (%)	15.520	–	17.303
Methionine total	0.240	–	0.361
Lysine total	0.620	–	0.904
Calcium (%)	20.001	–	4.091
Available phosphorus (%)	0.330	–	0.642

[1]Vitamin enrichment per kilogram of feed, vitamin A: 6,875 IU; vitamin D3: 1,640 IU; vitamin E: 2,000 IU; vitamin K3: 400 mg; biotin: 1.67 mg; thiamine: 83.3 mg; riboflavin B2: 1,000 mg; pyridoxine: 50 mg; vitamin B12: 2,000 mg; niacin: 6,666 mg; pantothenic acid: 2,000 mg; folic acid: 33.3 mg; antioxidant: 1,000 mg.

[2]Mineral enrichment per kilogram of feed, Cu: 10 mg; Fe: 60 mg; Mn: 100 mg; Zn: 80 mg; I: 233 mg; Se: 60 mg.

Evaluated Parameters

At 24, 72, 96, and 144 h after the challenge with *Salmonella* Enteritidis, 5 birds from each replicate (100 birds per treatment) were euthanized and necropsied, and samples of the crop, liver, and cecum were collected for quantification of *Salmonella* Enteritidis.

Quantification of Salmonella Enteritidis in The Crop, Liver and Cecum

The crop, cecum, and liver were removed and placed separately in sterile plastic bags, weighed, and macerated with a spatula. The amount of PBS added to achieve a ratio of 1:9 and a 10^{-1} dilution was determined according to the organ weight and content. After homogenization of the contents, a 1 mL aliquot was removed to generate the remaining dilutions (up to 10^{-8}) in test tubes containing 9 mL of PBS. A 0.1 mL aliquot of each dilution in duplicate was plated in BGA Nal/Rif and cultured for 24 h at 40°C. The number of CFUs per gram of content and organ was $\log_{10}$ transformed to interpret the results.

Experiment II

The experiment was conducted at FMVZ—UNESP, Botucatu Campus, Sao Paulo, Brazil at the facilities of the aviculture sector of Edgardia Experimental Farm, which belongs to the Department of Animal Production, and at the Laboratory of Avian Pathology of the Department of Clinical Veterinary Science.

Chickens and Facilities

A total of one hundred eighty 79-wk-old commercial Lohmann LSL-Lite laying hens housed in a bioclimatic chamber composed of 3 rooms were evaluated. Each room was equipped with 8 metal cages (1.00 m long × 0.45 m deep × 0.40 m high) arranged in 2 double rows with a central corridor. The cages were equipped with a nipple drinker and trough feeder placed at the front of the cage.

Sanitary Management

The vaccine program started in the hatchery and was maintained throughout the birds' lives as shown in Table 1.

Experimental Design

The experiment lasted 140 D, and the treatments were applied during the first 14 D of the experimental period. The experimental design was completely randomized with 3 treatments and 6 replicates with 10 birds each: group 1—wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum for 14 D, followed by production feed; group 2—fasting (California method) for 14 D, followed by production feed; group 3–production feed ad libitum (without forced molting).

During the forced molting period (28 D), no artificial light was provided. Only the birds in the control group (group 3) received 17 h of light per day, whereas the others were subjected to a natural photoperiod equivalent to 11 h. After the forced molting period, an extra 30 min of light was added per week, until a total of 17 h of light per day. The experiments were conducted during the spring period.

The birds in groups 1 and 2 that lost 25% of their body weight prior to the 14-D period were immediately fed production feed: on the first, second, and third days, 30, 60, and 90% of the amount that the birds would consume was supplied ad libitum; and from the fourth day on, the birds received the amount of feed recommended by the line manual until the 28th day of the experimental period. The subsequent 112 D were divided into 4 periods of 28 D to evaluate egg quality and performance.

Water was supplied ad libitum throughout the experimental period.

Evaluated Parameters—Zootechnical Performance

After the forced molting period, the following performance characteristics were evaluated.

Mean Chicken Weight Each of the birds belonging to the 3 treatments was weighed at 0, 7, 10, 14, 28, and 140 D of the experimental period. The weight of each bird was added to the weight of the other birds of each treatment and divided by the number of birds, thus obtaining the mean bird weight of each treatment.

Feed Intake (Grams) The feed was weighed weekly, and intake was obtained weekly by subtracting the leftover feed from the total provided and dividing the result by the mean number of birds, which was expressed in g/bird/d.

Laying Percentage (%) Calculated daily by dividing the number of eggs produced in the plot by the mean number of birds.

Egg Quality Assessed at the end of each 28-D period over 3 consecutive days using 1 egg per replicate (6 eggs per treatment), for a total of 18 eggs per day, or 54 eggs per 28-D cycle.

Percentage of Whole Eggs (%)

Mean Egg Weight (Grams) Calculated on a weekly basis by dividing the weight of the eggs of each plot by the number of eggs.

Egg Mass (Grams)

Feed Conversion Per Egg Mass Calculated on a weekly basis by dividing feed intake by egg mass.

Feed Conversion Per Dozen Eggs Calculated on a weekly basis by dividing the feed intake by the laying rate and multiplying the result by 12 (one dozen).

Viability

Specific Gravity

Shell Breaking Strength Obtained via evaluations in a texturometer (TA.XTPlus Texture Analyzer) equipped with a 75 mm probe (p/2) and a test speed of 1 mm/s.

Shell, Yolk, and Albumen Percentages

Shell Weight Per Unit Surface Area Obtained by the following formula:

$$\mathrm{SWUSA} = \{\mathrm{SW}/[3.9782\,\mathrm{x}\,(\mathrm{EW}\,0.7056)]\}\,\mathrm{x}\,1000,$$

where SW = shell weight and EW = egg weight, expressed in mg/cm^2.

Shell Thickness Obtained by the mean of 3 measurements obtained at the shell equator using a digital caliper (0.01 mm accuracy).

Haugh Unit Calculated using the following formula:

$$\mathrm{HU} = 100\log(\mathrm{H} + 7.57 - 1.7\ \mathrm{W}^{\wedge}0.37),$$

where H = albumen height (mm) and W = egg weight (g) according to Card and Nesheim (1968).

Quantification of Serum Antibodies Against Newcastle Disease Virus

At 14, 28, 84, and 140 D after the beginning of forced molting, 3 mL of blood was collected by puncturing the brachial vein of 12 hens per experimental group. The samples were stored in microtubes and then centrifuged under refrigeration (1,500 × g, 5 min, 8°C) to obtain serum, which was stored at –80°C until analysis.

Detection of antibodies against Newcastle disease virus was performed using a commercial ELISA kit (IDEXX NDV Ab Test for chickens, IDEXX Laboratories) according to the manufacturer's instructions. The assay was performed to quantify the production of antibodies against Newcastle disease virus.

Quantification of Plasma Corticosterone Levels

One day before the beginning of forced molting and one day after the end of the molting period, 3 mL of blood was collected by puncturing the brachial vein of 12 hens per experimental group. To reduce stress and subsequent effect of stress on plasma corticosterone levels, the samples were collected in the morning before sunrise, stored in microtubes, and then centrifuged under refrigeration (3,000 × g, 5 min, 8°C) to obtain the plasma, which was stored at –80°C until analysis.

The plasma corticosterone levels were determined using a commercial ELISA kit (DetectX® Corticosterone Immunoassay Kit, Arbor Assays, MI) according to the

Table 3. Crop and ceca colonization by *Salmonella* Enteritidis (SE) in commercial laying hens challenged with 10^8 CFU/mL of SE on the fourth day after the start of forced molting.

Treatment[1]	Crop—CFU $\log_{10}$/g ± SEM[2] Hours after the challenge with SE 24	72	96	144
Wheat bran	3.1 ± $0.55^{b\ A}$	2.4 ± $0.81^{b\ A,B}$	0.72 ± $0.12^{b\ B}$	0^{C}
California method	6.6 ± $1.1^{a\ A}$	6.1 ± $0.56^{a\ A}$	6.0 ± $0.88^{a\ A}$	0^{B}
Ad libitum feed	0.6 ± $0.22^{c\ A}$	$0^{c\ B}$	0.78 ± $0.13^{b\ A}$	0^{B}
Wheat bran	5.3 ± $0.56^{a\ A,B}$	6.8 ± $0.58^{a\ A}$	4.4 ± $0.98^{a,b\ A,B}$	4.3 ± $0.90^{a\ B}$
California method	6.9 ± $0.32^{a\ A,B}$	8.1 ± $0.29^{a\ A}$	5.9 ± $0.21^{a\ A,B}$	5.8 ± $0.54^{a\ B}$
Ad libitum feed	4.8 ± 0.77^{a}	3.6 ± 0.88^{b}	2.6 ± 0.43^{b}	3.7 ± 0.33^{a}

[a,b]Different letters in the column and [A, B]different letters in the line differ by the Tukey test ($P < 0.05$).
[1]Group 1: wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 10 days; group 2: total feed restriction (California method) during 10 D; group 3: production feed ad libitum without any type of restriction.
[2]SEM: standard error of the mean.

manufacturer's instructions. The plasma corticosterone concentration was determined by a standard curve and expressed in picograms per milliliter of plasma corticosterone.

Statistical Analysis

The results of the *Salmonella* Enteritidis count were log-transformed to reach normality and subsequently subjected to analysis of variance followed by Tukey's test for multiple comparisons (5%). The mean and standard error (±SEM) were also calculated.

The other data were subjected to an analysis of variance, and differences between treatments were compared by Tukey's test at 5%. The data were analyzed using Mixed procedure in the SAS 9.1 software (SAS Institute, 1993).

RESULTS AND DISCUSSION

Experiment I

Forced molting is commonly used to prepare hens for a second egg production cycle. This process is nothing more than an imitation of what happens in nature, where birds enter the incubation phase to reproduce. During this period, involution of the reproductive system occurs, followed by a pause in egg production and feather loss and replacement (Sherry et al., 1980; Ricke et al., 2013). During molting, the hen voluntarily reduces food intake and can lose up to 20% of her body weight, which is a process called spontaneous anorexia (Mrosovsky and Sherry, 1980; Ricke et al., 2013). After this process, the hen resumes feed intake and starts a new reproductive cycle (Sherry et al., 1980).

Over the years, researchers have developed artificial molt induction methods, which result in a pause in egg production and feather loss. These results are obtained by withdrawing feed, water, or feed and water and reducing the daily photoperiod (Ricke et al., 2013). Thus, the reproductive life of commercial laying hens can be extended from less than 80 wk to more than 110 wk (Bell, 2003). For a long time, total feed withdrawal was the most commonly used method for inducing forced molting because of its ease of implementation, and it results in weight loss and satisfactory hen performance after this period. However, hen welfare and health issues have led researchers to seek less severe options for implementing molting (Ricke et al., 2013).

With regard to hen health, forced molting has been shown to significantly affect infection by *Salmonella* Enteritidis at different times of the infection cycle (Holt, 1993; Holt and Porter, 1993). Natural infection by *Salmonella* mainly occurs orally in birds via colonization of digestive system organs, such as the crop and ceca, which are the main colonization sites (Impey and Mead, 1989; Andreatti Filho et al., 1997, 2000). In our study (Table 3) we found a strong influence of fasting on the colonization of *Salmonella* Enteritidis. The hens that underwent total fasting (group 2) presented higher contamination ($P < 0.05$) of *Salmonella* Enteritidis as evidenced by the crop analysis (3.5, 3.7, and 5.2 log units at the 24, 72, and 96 h after the challenge, respectively, when compared to the hens from group 1 that received as alternative diet). According to Corrier et al. (1999), chickens subjected to feed fasting present exacerbated crop colonization by *Salmonella* Enteritidis, and in our study the level of contamination remained high until 96 h after infection. However, the bacteria could not be isolated at 144 h after infection, which does not mean that it was no longer present in the organ because molting can cause a recurrence of infection (Holt and Porter, 1993).

The success in colonizing the crop seems to be a determinant of the intestinal invasion and internal organ colonization by *Salmonella* Enteritidis (Ricke, 2003). In our study, although the hens that were subjected to feed fasting (group 2) and supplied the alternative diet (group 1) presented high crop colonization, the bacterium could not be isolated in the liver at any of the times studied (data not shown). This finding is inconsistent with the claim of Durant et al. (1999), who showed that invasion of the spleen and liver of hens subjected to fasting was increased by at least 5 times compared with that of nonfasted hens. Nakamura et al. (2004) showed the use of vaccination against *Salmonella* Enteritidis before forced molting can reduce the bacterial counts.

Table 4. Mean body weight of commercial laying hens submitted to different molt induction programs.

Treatment[1]	Induced molting period (days)/weight (g) ± SEM[2]					
	0	7	10	14	28	140
Wheat bran	1.735 ± 30.5	1.423 ± 28.5[B]	1.405 ± 33.9[B]	1.439 ± 18.5[B]	1.616 ± 17.5[B]	1.724 ± 43.1
California method	1.751 ± 38.2	1.353 ± 22.4[C]	1.300 ± 23.5[C]	1.403 ± 34.7[B]	1.611 ± 40.9[B]	1.717 ± 23.7
Ad libitum feed	1.743 ± 33.6	1.739 ± 27.9[A]	1.724 ± 31.7[A]	1.733 ± 32.8[A]	1.716 ± 42.8[A]	1.745 ± 48.8

[A,B]Means followed by different letters in the column differ statistically by the Tukey test ($P < 0.05$).
[1]Group 1: wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 14 D, followed by production feed; group 2: total feed restriction (California method) during 14 D, followed by production feed; group 3: feed ad libitum without any type of restriction.
[2]SEM: standard error of the mean.

According to Durant et al. (1999), a 9-D fasting period is able to increase the cecal colonization by *Salmonella* Enteritidis by more than 3 log units compared to hens not subjected to any type of feed restriction. In our study, at 24 h after infection, we found no difference ($P < 0.05$) in the quantification of cecal colonization by *Salmonella* Enteritidis in the different groups evaluated, even when compared to hens that did not undergo any type of feed restriction (group 3). However, in the periods of 72 and 96 h after infection, the hens that did not undergo any type of feed restriction presented lower ($P < 0.05$) cecal colonization by the bacterium. No differences were found between the hens that had fasted and received the alternative diet. Dunkley et al. (2007) observed a reduction in the amount of *Salmonella* Enteritidis in the feces and ceca of hens fed alfalfa crumbles compared to hens subjected to forced molting (California method).

Previous studies have established that feed fasting changes the microenvironment of the crop and ceca and increases the susceptibility of hens to microorganism colonization and systemic infections, especially by *Salmonella* Enteritidis in the case of commercial laying hens (Ricke et al., 2013). Therefore, several alternative forced molting methods have been investigated with the aim of maintaining hen well-being and intestinal microbiota balance while also achieving body weight loss and reproductive apparatus rest, which are necessary for the beginning of a new egg production cycle.

Alternative forced molting methods include nutritionally imbalanced diets, low-palatability diets, hormone administration, or low-energy diets (Bell, 2003; Holt, 2003; Park et al., 2004). In search of alternative forced molting methods, Molino et al. (2009) observed that wheat bran provided ad libitum yielded the same productive performance as the conventional method (fasting), had a low cost, and was easy to apply. Santos et al. (2014) also demonstrated that wheat bran and rice bran yielded a better performance in terms of egg production and quality, thus providing better welfare for hens subjected to forced molting. Biggs et al. (2003) found that wheat bran-based diets were effective for inducing molting without the fasting period and yielded good egg production in the second laying cycle. Gongruttananum and Saengkudrua (2016) treated hens with cassava meal as a dietary alternative for forced molting and did not observe differences with the control except for an increase in the albumen weight and mortality rate. In our study, we used a mixture of wheat bran and coarse limestone (70 and 30%, respectively) offered ad libitum to hens as an alternative forced molting method. Intake of the mixture appears to have been effective in helping control crop colonization by *Salmonella* Enteritidis, although a reduction of ceca colonization was not observed. Although the issue of infection by *Salmonella* Enteritidis is of great importance when inducing forced molting in birds, this issue should not be evaluated in isolation as a criterion for selecting a forced molting method.

Experiment II

In Experiment II, we evaluated production and egg quality indexes and measured an indicator of well-being and the efficiency of the immune system.

The forced molting process translates into a period of physiological rest and consequent rejuvenation of the reproductive system, and it is associated with significant weight loss and fat accumulation reduction (Cunningham and Mauldin, 1996). In this study, before the beginning of the forced molting period, the hens had a mean weight of 1,743 g. The mean body weight of the hens over the 140 D of the experiment (before and after forced molting) is shown in Table 4.

On the seventh and tenth days after the beginning of forced molting, the hens of the control group (group 3) presented higher ($P < 0.05$) body weight compared to the hens that underwent forced molting, regardless of the method implemented. At 7 D, hens that were fasted experienced weight loss of 22.73% relative to the initial weight, and hens that received wheat bran experienced weight loss of 17.98%. At 10 D, the fasted hens lost 25.76% of their body weight, whereas those that received wheat bran lost 19.02%. After the 10th day, the hens subjected to fasting were again supplied the production feed.

On day 14, the hens that were not subjected to any type of feed restriction (group 3—control), compared to the hens of the other groups, presented the highest ($P < 0.05$) body weights (1.733 g). The fasted hens presented higher weights than at earlier time periods because they had resumed feeding as previously mentioned. At 14 D, these hens had lost 17.06% of their body weight and

Table 5. Performance of commercial laying hens submitted to different molt induction programs during the second production cycle.

Variables	Treatment[1]		
	Wheat bran	California method	Ad libitum feed
Feed intake (g/bird/d)	116.16 ± 8.2[2]	117.70 ± 10.2	120.38 ± 7.2
Viability (%)	93.75 ± 0.31	91.25 ± 0.23	91.43 ± 0.38
Egg production rate (%)	89.15 ± 3.2^{A}	82.73 ± 2.1^{B}	81.75 ± 3.9^{B}
Viable egg rate (%)	98.85 ± 0.11	98.74 ± 0.22	98.82 ± 0.02
Egg weight (g)	67.16 ± 0.46	67.81 ± 1.44	66.35 ± 2.11
Egg mass (g)	59.80 ± 4.2^{A}	56.09 ± 3.3^{B}	54.22 ± 4.3^{B}
Feed conversion/kg eggs	1.96 ± 0.03^{B}	2.10 ± 0.01^{A}	2.22 ± 0.13^{A}
Feed conversion/dozen eggs	1.58 ± 0.12^{B}	1.71 ± 0.01^{A}	1.77 ± 0.03^{A}

^{A,B}Means follow by different letters in the line differ statistically by the Tukey test ($P < 0.05$).
[1]Group 1; wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 14 D, followed by production feed; group 2: total feed restriction (California method) during 14 D, followed by production feed; group 3: feed ad libitum without any type of restriction.
[2]SEM: standard error of the mean.

Table 6. Egg quality evaluation of commercial laying hens submitted to different molt induction programs during the second production cycle.

Variables	Treatment[1]		
	Wheat bran	California method	Ad libitum feed
Specific gravity (g/cm^3)	1.083 ± 4.13^{A}[2]	1.084 ± 7.18^{A}	1.080 ± 8.33^{B}
Shell breaking strength (gf)	3.248 ± 1.13^{A}	3.472 ± 5.22^{A}	2.981 ± 8.23^{B}
Shell percentage (%)	8.46 ± 1.55^{A}	8.59 ± 4.03^{A}	8.26 ± 0.88^{B}
Shell weight per unit surface area (mg/cm^2)	74.56 ± 9.09^{A}	75.91 ± 7.55^{A}	72.39 ± 5.45^{B}
Shell thickness (mm)	0.39 ± 0.05^{A}	0.39 ± 0.04^{A}	0.38 ± 0.01^{B}
Yolk percentage (g)	26.06 ± 2.35	26.51 ± 0.95	26.98 ± 3.05
Albumen percentage (%)	65.48 ± 1.02	65.23 ± 2.15	64.75 ± 1.99
Haugh unit	92.26 ± 2.13^{A}	93.86 ± 4.73^{A}	86.97 ± 8.13^{B}

^{A,B}Means follow by different letters in the line differ statistically by the Tukey test ($P < 0.05$).
[1]Group 1: wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 14 D, followed by production feed; group 2: total feed restriction (California method) during 14 D, followed by production feed; group 3: feed ad libitum without any type of restriction.
[2]SEM: standard error of the mean.

were again provided ad libitum access to production feed.

At 28 D, hens subjected to forced molting had not yet recovered their body weight ($P < 0.05$) compared to the hens of the control group. At the end of the experiment (140 D), the hens of all experimental groups presented similar body weights within the range recommended by the Lohmann LSL-Lite line manual (Lohmann do Brasil, 2017), thus demonstrating the total recovery of the hens that were subjected to forced molting.

According to Table 5, the molting methods did not influence the feed intake, percentage of viable eggs, egg weight, or viability during the second production cycle.

In relation to the other performance characteristics (egg production rate, egg mass, and feed conversion per dozen and egg mass), the hens that received wheat bran (group 1) presented higher results ($P < 0.05$) compared to the other experimental groups, and these hens also presented a better performance than the hens subjected to molting by the traditional method of feed fasting. These results show that weight loss of at least 25%, which is based on recommendations in the literature (Buhr and Cunninghan, 1994), is not necessary to achieve good performance during the second production cycle. However, contradictory results have been obtained regarding the effect of weight loss on post-molt performance, with some studies concluding that weight loss of approximately 25 to 35% is necessary for maximum production post-molt (Brake et al., 1981; Baker et al., 1983; McDaniel, 1985) and others reporting that weight loss during molting has little effect on post-molt performance (Cunningham and McCormick, 1985; Zimmermann et al., 1987; Zimmermann and Andrews, 1990).

According to Table 6, the different molting methods did not have an effect ($P < 0.05$) on egg quality except for the yolk and albumen percentage. The Haugh unit and all egg shell quality characteristics were superior ($P < 0.05$) in the eggs from hens that had undergone forced molting (both wheat bran and fasting) compared to the eggs from hens that had not undergone forced molting, which was likely a result of the rest and regeneration of the reproductive system that occurs with forced molting. Thus, hens that were subjected to fasting or that received wheat bran presented improvements in egg shell quality. Other studies have also reported improvements in egg characteristics from hens subjected to forced molting, such as increased weight, size, and shell strength (Aygun, 2013; Mitrović et al., 2016; Shaker and Kirkuki, 2017). Moreover, molting via

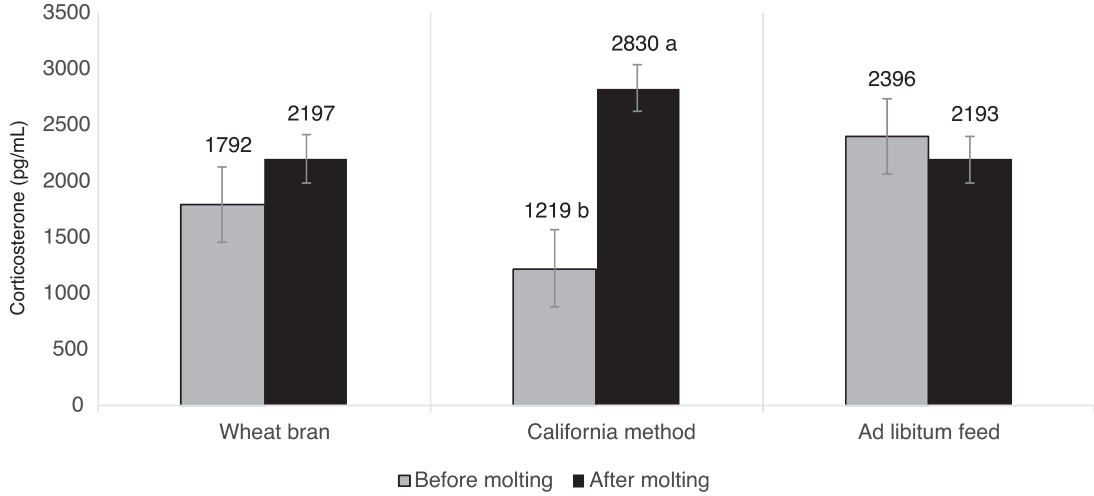

Figure 1. Plasma corticosterone levels (pg/mL) before and after the forced molting period in commercial laying hens submitted to different molt induction programs. [a, b]Means followed by different letters in the bars differ statistically by the Tukey test ($P < 0.05$). Group 1: wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 14 D, followed by production feed; group 2: total feed restriction (California method) during 14 D, followed by production feed; group 3: feed ad libitum without any type of restriction.

alfalfa meal and barley grain may be used as nonfeed removal programs because it does not produce negative effects on the performance and egg quality parameters of brown layers (Sariozkan et al., 2016).

In addition to production and egg quality indexes, the issue of animal welfare should be considered when choosing a forced molting method. The physiological parameters of stress in production poultry may include increased corticosterone levels and suppression of the humoral response (Sandhu et al., 2007; Golden et al., 2008). Corticosterone is the major glucocorticoid released by the adrenal gland, and increased levels of corticosterone in the plasma or serum are considered an indicator of stress in chickens (Gross and Siegel, 1983). The release of high amounts of corticosterone may cause the involution of lymphoid tissue (spleen, thymus, and bursa of Fabricius), suppress humoral and cellular immunity, and induce metabolic and behavioral changes (El-Lethey et al., 2003).

In our study, we compared the plasma corticosterone levels before and after the forced molting period and found that the hens subjected to feed fasting (group 2) showed elevated ($P < 0.05$) hormone levels (56.9%) compared with the pre-molt levels (Figure 1). Hens receiving an alternative diet (group 1) did not present a significant increase in plasma corticosterone levels, although an increase of 18.3% was observed. In the control group (group 3), which was represented by the hens that received feed ad libitum, a numerical reduction in corticosterone levels was observed, which was likely caused by adaptation of the animals to the environment. Onbaşilar and Erol (2007) evaluated hens subjected to feed fasting as part of the conventional forced molting method and also found high levels of plasma corticosterone compared with that of hens fed diets containing barley and zinc.

According to Golden et al. (2008), prolonged fasting is a potent stressor imposed on commercial layers that raises plasma corticosterone levels, thereby altering the count and functional activity of leukocytes, including T and B lymphocytes, which may impair the immune response of hens and reduce the antibody response against an antigen (Gross et al., 1980).

In our study, we evaluated antibody titers against the Newcastle disease virus at 14, 28, 84, and 140 D of the second cycle (Figure 2). In the first evaluation at 14 D, we observed that all hens (groups 1, 2, and 3) presented similar and satisfactory vaccine titers, which was expected based on the vaccine history of the flock (Table 1). At 28 D, the hens subjected to the alternative diet (group 1) presented lower ($P < 0.05$) vaccine titers than the hens of the other experimental groups. At 84 D, hens in the control group (group 3), which were not subjected to forced molting, had higher ($P < 0.05$) vaccine antibodies titers compared with the hens in the other experimental groups. At 140 D, hens in the fasted group presented lower ($P < 0.05$) titers compared with the hens of the other groups (groups 1 and 3). In addition, during the experimental period, the vaccine antibody titers declined in hens of all experimental groups.

Respiratory diseases are among the major problems in poultry production, and Newcastle disease virus is one of the most commonly reported respiratory pathogens (Kapczynski et al., 2013). Sandhu et al. (2006) showed that by inducing forced molting through fasting, the titers of antibodies against Newcastle disease virus decreased. The authors believe that this drop is due to the effects of increased plasma corticosterone levels.

The use of wheat bran for forced molting is feasible because a reduction in *Salmonella* Enteritidis was observed in the crop and a better performance (egg production, egg weight, and feed conversion/kg per dozen eggs) and lower corticosterone production were observed compared with that of hens subjected to

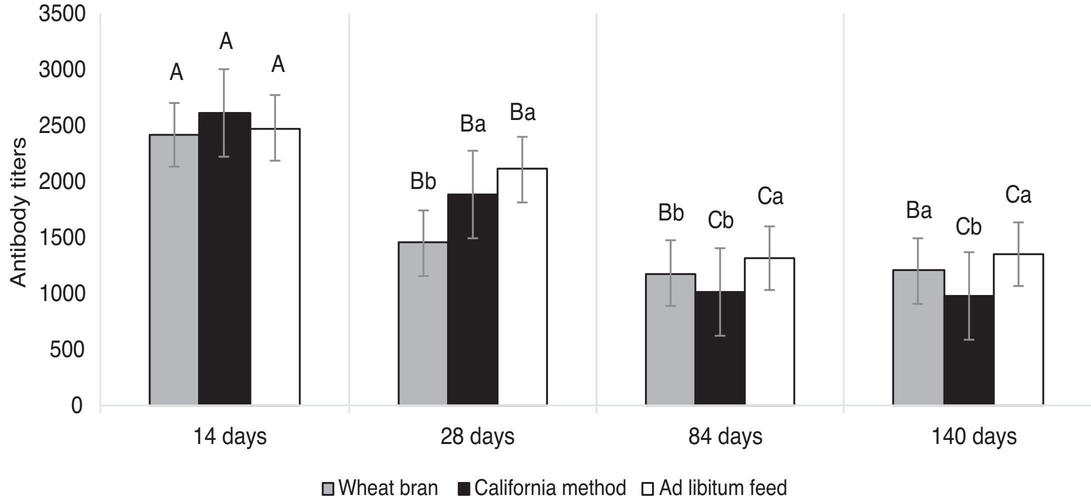

Figure 2. Antibody titers against the Newcastle disease virus in commercial laying hens submitted to different molt induction programs during the second production cycle. [A, B]Means followed by different upper case letters in the bars (in different periods) and [a, b] means followed by different lower case letters in the bars (in the same period) differ statistically by the Tukey test ($P < 0.05$). Group 1: wheat bran-based diet with coarse limestone (70 and 30%, respectively) offered ad libitum during 14 D, followed by production feed; group 2: total feed restriction (California method) during 14 D, followed by production feed; group 3: feed ad libitum without any type of restriction.

total fasting. Thus, using wheat bran for forced molting represents a practical alternative that better meets the welfare needs of hens.

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